

DESIGN AND SIMULATION OF DYNAMICS AND CONTROL SYSTEMS FOR A SINGLE FIXED WING UAV



DYSL-AT (DRDO YOUNG SCIENTIST LABORATORY-ASYMMETRIC TECHNOLOGIES)

INTERNSHIP PROJECT REPORT

Submitted in partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

AERONAUTICAL ENGINEERING

Undertaken at

DYSL-AT, DRDO

HYDERABAD

Submitted by

AMARANENI VINITHA

Supervised by

MR. V V S M S GANESH, Scientist-C

DYSL-AT, DRDO, HYDERABAD



HINDUSTAN
INSTITUTE OF TECHNOLOGY & SCIENCE
(DEEMED TO BE UNIVERSITY)

DEPARTMENT OF AERONAUTICAL ENGINEERING

HINDUSTAN INSTITUTE OF TECHNOLOGY AND SCIENCE

PADUR, CHENNAI – 603 103

MARCH – 2025

ABSTRACT

This project presents the design and simulation of the dynamics and control systems for a single fixed-wing unmanned aerial vehicle (UAV). A six-degree-of-freedom (6-DOF) nonlinear mathematical model was developed in MATLAB based on Newton–Euler rigid-body equations of motion. The model incorporates aerodynamic forces and moments, propulsion effects, gravitational forces, and control surface inputs to accurately represent the aircraft dynamics.

A trim analysis was performed to determine steady-level flight conditions at a nominal airspeed of 25 m/s. The nonlinear model was subsequently linearized about the trim condition to obtain longitudinal and lateral-directional state-space representations. Transfer function models were derived and used for control system development.

A hierarchical autopilot architecture employing proportional–integral–derivative (PID) controllers was designed for roll, course, pitch, altitude, airspeed, and sideslip regulation. Controller gains were computed using standard aircraft control design methodologies and validated through closed-loop simulations.

Simulation results demonstrate successful stabilization, command tracking, and stable operation under Dryden wind-disturbance conditions. For an airspeed command change from 25 m/s to 28 m/s, the system achieved a settling time of approximately 6.24 s with a maximum overshoot of 0.27%. The final steady-state airspeed error was 0.083 m/s. Altitude tracking was maintained with an average error of 10.07 m, while course tracking achieved an average error of 14.22°. The developed control system successfully maintained stable flight while achieving satisfactory airspeed, altitude, and course tracking performance under simulated operating conditions.

The developed framework provides a foundation for advanced UAV flight control research and can be extended to incorporate robust, adaptive, and autonomous control methodologies in future work.

CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

Unmanned Aerial Vehicles (UAVs) have become increasingly important in modern aerospace applications due to their versatility, cost-effectiveness, and ability to operate in environments that may be inaccessible or hazardous to human pilots. UAVs are widely used in surveillance, environmental monitoring, disaster management, precision agriculture, infrastructure inspection, and scientific research. The growing demand for autonomous and reliable UAV operations has increased the need for accurate flight dynamic models and robust flight control systems.

A UAV is a highly dynamic system whose behavior is governed by complex interactions between aerodynamic forces, propulsion systems, and control surface inputs. To ensure stable and efficient flight, it is essential to understand the vehicle's dynamic characteristics and develop appropriate control strategies capable of maintaining desired flight conditions.

This project focuses on the development of a complete six-degree-of-freedom (6-DOF) flight dynamics model of a fixed-wing UAV based on the Aerosonde aircraft configuration. The nonlinear aircraft model is implemented in MATLAB and used to perform trim analysis, linearization, transfer function extraction, and autopilot design. A hierarchical PID-based autopilot architecture is then developed to regulate key flight variables such as airspeed, altitude, pitch attitude, roll attitude, course angle, and sideslip angle.

The performance of the developed control system is evaluated through nonlinear simulation studies under commanded flight conditions and wind disturbances. Various airspeed, altitude, and course commands are applied to assess the aircraft's tracking capability, stability characteristics, disturbance rejection capability, and overall control performance.

The project provides a systematic framework for understanding UAV flight dynamics and serves as a foundation for more advanced control techniques that may be explored in future work.

1.2 Problem Statement

Fixed-wing UAVs exhibit highly coupled nonlinear dynamics and are subject to variations in aerodynamic forces, vehicle states, and environmental conditions. Without an effective control system, maintaining stable flight and accurately following commanded trajectories becomes difficult.

The design of a reliable autopilot requires accurate mathematical modeling of the aircraft dynamics, determination of steady-state trim conditions, derivation of linearized models suitable for control design, and implementation of control algorithms capable of regulating the vehicle during flight.

Therefore, the primary challenge addressed in this project is the development of a flight control system that enables a fixed-wing UAV to maintain stable flight and accurately track commanded airspeed, altitude, and course commands. This requires integrating flight dynamics modeling, trim computation, transfer function development, and PID controller design into a unified simulation framework.

1.3 Objectives

The main objectives of this project are:

1. To develop a nonlinear six-degree-of-freedom (6-DOF) mathematical model of a fixed-wing UAV using MATLAB.
2. To simulate the longitudinal and lateral-directional dynamics of the aircraft under different operating conditions.

3. To determine steady-state trim conditions corresponding to a specified airspeed and flight-path angle.
4. To linearize the nonlinear aircraft model around the trim condition and obtain state-space representations suitable for control system design.
5. To derive transfer function models representing the aircraft's longitudinal and lateral dynamic behavior.
6. To design and implement a hierarchical PID-based autopilot architecture consisting of:
 - Roll hold controller
 - Course hold controller
 - Pitch hold controller
 - Altitude hold controller
 - Airspeed hold controller
 - Sideslip regulation controller
7. To evaluate controller performance using simulation-based metrics such as:
 - Settling time
 - Overshoot
 - Steady-state tracking error
 - Command-following capability

8. To validate the effectiveness of the developed autopilot through nonlinear closed-loop simulations using altitude, airspeed, and course commands.
9. To establish a simulation framework that can be extended in future work to incorporate advanced control approaches such as adaptive control, optimal control, and model predictive control.

1.4 Scope of the Project

This project focuses on the modeling, simulation, and control of a fixed-wing UAV using MATLAB-based aerospace simulation tools. The work includes the development of a nonlinear aircraft dynamics model, trim condition analysis, linearization of the nonlinear system, derivation of transfer function models, and implementation of a PID-based autopilot system.

The study utilizes the Aerosonde UAV configuration and assumes known aerodynamic and physical parameters. The developed simulation framework considers the aircraft's translational and rotational dynamics together with atmospheric wind disturbances represented using the Dryden turbulence model.

The project scope includes:

- Six-degree-of-freedom flight dynamics modeling
- Aircraft trim computation
- Longitudinal and lateral linearization
- Transfer function derivation
- PID controller design and tuning
- Closed-loop autopilot implementation

- Performance evaluation through simulation

The project does not include hardware implementation, real-time flight testing, advanced state estimation techniques, adaptive control systems, or model predictive control. These topics are considered potential areas for future research and development.

CHAPTER 2

THEORETICAL BACKGROUND AND RELATED WORK

2.1 Introduction

Unmanned Aerial Vehicles (UAVs) have become an important research area due to their increasing use in surveillance, reconnaissance, environmental monitoring, and autonomous transportation applications. The development of reliable flight control systems requires accurate modelling of aircraft dynamics, trim analysis, system linearization, and controller design. Fixed-wing UAVs present unique challenges because of the coupling between longitudinal and lateral-directional dynamics and the presence of aerodynamic nonlinearities.

To achieve stable autonomous flight, researchers commonly employ mathematical modelling techniques based on six-degree-of-freedom rigid-body dynamics. Once a nonlinear model is developed, trim conditions are determined and the model is linearized around an operating point to facilitate control system design.

2.2 Aircraft Dynamic Modelling

Aircraft dynamic modelling forms the foundation of flight control system development. The motion of a fixed-wing aircraft is described using six-degree-of-freedom equations consisting of three translational and three rotational motions.

The complete nonlinear model includes:

- Aerodynamic forces and moments
- Gravitational forces
- Propulsion effects

- Control surface deflections

These equations are typically derived using Newton-Euler rigid-body formulations and implemented in simulation environments such as MATLAB/Simulink. These mathematical models provide the foundation for trim analysis, system linearization, and controller design presented in the subsequent sections.

Accurate dynamic models allow engineers to predict aircraft behaviour under different flight conditions and serve as the basis for controller development.

2.3 Trim Analysis and Linearization

Trim analysis is used to determine steady-state operating conditions where all forces and moments acting on the aircraft are balanced.

For a fixed-wing UAV, trim conditions provide:

- Equilibrium airspeed
- Angle of attack
- Pitch attitude
- Elevator deflection
- Throttle setting

Once a trim condition is established, the nonlinear model can be linearized around the operating point.

Linearized models are advantageous because they:

- Simplify control design

- Allow transfer function derivation
- Enable stability analysis
- Support controller tuning

The resulting state-space representation provides insight into both longitudinal and lateral-directional aircraft dynamics.

2.4 Flight Control Systems

Flight control systems are responsible for maintaining aircraft stability and tracking desired commands.

Common control approaches include:

PID Control

PID Controller Equation:

$$u(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{de(t)}{dt}$$

where $u(t)$ is the controller output, $e(t)$ is the tracking error, and K_p , K_i , and K_d represent the proportional, integral, and derivative gains, respectively.

PID controllers remain the most widely used control strategy because of their:

- Simplicity
- Ease of implementation
- Low computational requirements
- Robust performance

PID controllers are commonly employed for:

- Roll control
- Pitch control
- Altitude hold
- Airspeed regulation
- Course tracking

Advanced Control Methods

Several advanced techniques have been proposed, including:

- Linear Quadratic Regulator (LQR)
- Model Predictive Control (MPC)
- Adaptive Control
- Robust Control

Although these methods may provide improved performance, they typically require more complex mathematical models and greater computational resources.

2.5 Successive Loop Closure Autopilot Design

Successive loop closure is a widely adopted methodology for aircraft autopilot design.

The approach separates the control problem into multiple interconnected loops, as shown in Figure 2.1.

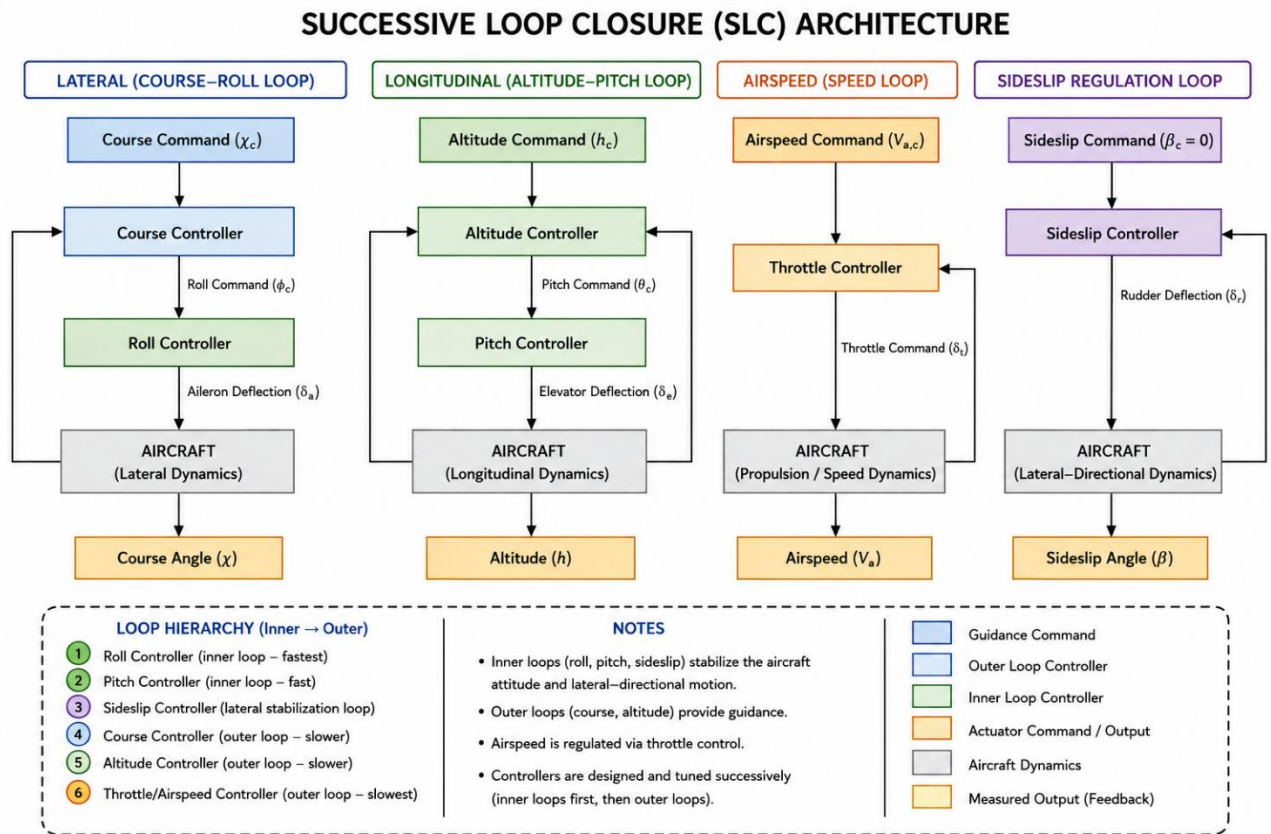


Figure 2.1: Successive Loop Closure (SLC) Autopilot Architecture for Fixed-Wing UAV Control

Figure 2.1 illustrates the hierarchical autopilot architecture developed in this project. The roll and pitch controllers form the inner stabilization loops, while the course and altitude controllers generate reference commands for the inner loops. Airspeed is regulated through throttle control, and sideslip is regulated through rudder deflection. The successive loop closure approach simplifies controller design and tuning by separating the control problem into multiple interconnected loops with different dynamic response characteristics.

Lateral Control

- Roll Hold
- Course Hold
- Sideslip Regulation

Longitudinal Control

- Pitch Hold
- Altitude Hold
- Airspeed Hold

The inner loops are designed first because they possess faster dynamics. Outer loops are then built around the stabilized inner loops.

This hierarchical structure simplifies controller tuning and improves system stability. In this project, the successive loop closure methodology was implemented using PID controllers for roll, course, pitch, altitude, airspeed, and sideslip regulation.

2.6 Summary

This chapter presented the theoretical background required for UAV flight control system development. The discussion covered aircraft dynamic modelling, trim analysis, linearization, PID control, and successive loop closure autopilot design. These concepts form the basis for the methodology adopted in this project and provide the framework for the modelling, simulation, and control system implementation presented in the following chapters.

CHAPTER 3

METHODOLOGY

3.1 Introduction

The objective of this project is to develop and evaluate an autopilot system for a fixed-wing Unmanned Aerial Vehicle (UAV) using classical control techniques. The methodology consists of aircraft dynamic modelling, trim analysis, model linearization, transfer function extraction, controller design, and simulation-based performance evaluation.

The complete UAV model was implemented in MATLAB using a six-degree-of-freedom nonlinear flight dynamics framework. The aircraft was first trimmed at a desired operating condition, after which the nonlinear model was linearized to obtain transfer functions and state-space models. Successive loop closure techniques were then employed to design PID-based autopilot controllers for roll, course, pitch, altitude, airspeed, and sideslip regulation.

The overall methodology adopted in this work is illustrated in Figure 3.1.

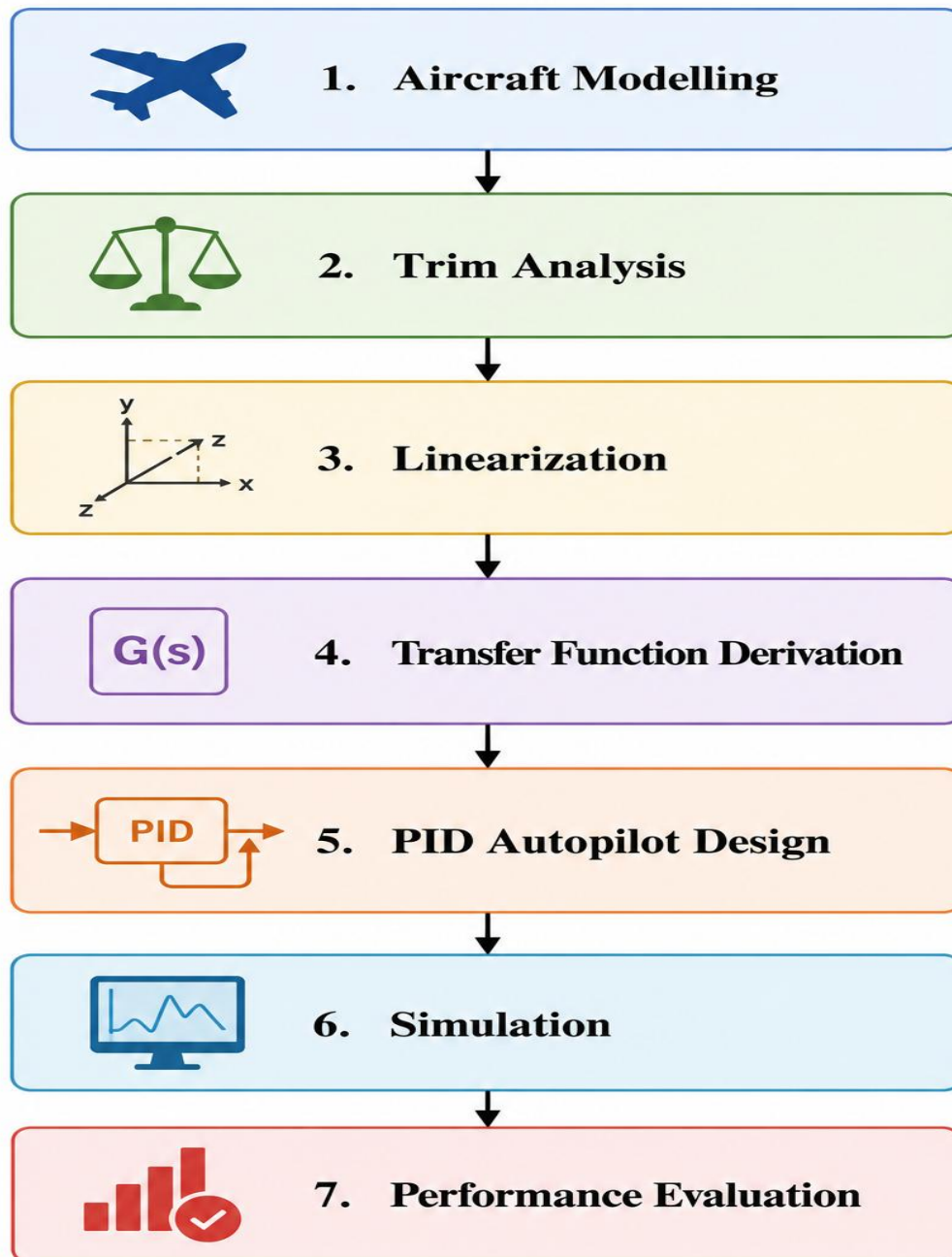


Figure 3.1: Methodology Flowchart for UAV Autopilot Development

Figure 3.1 presents the overall workflow adopted for the development of the UAV autopilot system. The process begins with nonlinear aircraft modelling, followed by trim analysis and model linearization. Transfer function models are then derived and used for PID autopilot design. Finally, the complete control system is evaluated through nonlinear simulations and performance analysis.

3.2 Aircraft Dynamic Modelling

The UAV was modeled as a rigid body with six degrees of freedom (6-DOF). The model incorporates translational and rotational motion in three-dimensional space and represents the complete nonlinear flight dynamics of the aircraft.

The state vector consists of:

$$x = [p_n, p_e, p_d, u, v, w, e_0, e_1, e_2, e_3, p, q, r]^T$$

where:

Variable	Description
p_n	North Position
p_e	East Position
p_d	Down Position
u, v, w	Body-axis velocities
e_0, e_1, e_2, e_3	Quaternion states
p, q, r	Angular rates

The control input vector is given by:

$$u = [\delta_e, \delta_t, \delta_a, \delta_r]^T$$

where:

Input	Description
δ_e	Elevator Deflection
δ_t	Throttle
δ_a	Aileron Deflection
δ_r	Rudder Deflection

The nonlinear equations of motion include:

- Aerodynamic forces and moments
- Propulsion effects
- Gravitational forces
- Wind disturbances

Wind disturbances were modeled using the Dryden turbulence model to provide a realistic atmospheric environment for controller evaluation. The aircraft parameters correspond to the Aerosonde UAV model provided in the simulation framework.

Numerical integration of the nonlinear equations was performed using a fourth-order Runge-Kutta algorithm. The resulting nonlinear aircraft model formed the basis for the subsequent trim analysis and controller design.

3.3 Trim Analysis

Trim analysis was performed to determine a steady-state flight condition in which all forces and moments acting on the aircraft are balanced.

For level flight, the trim condition satisfies:

$$\dot{x} = 0$$

subject to:

$$V_a = 25 \text{ m/s}$$

$$\gamma = 0$$

where:

Symbol	Description
V_a	Airspeed
γ	Flight Path Angle

The trim condition corresponds to steady-level flight, where the aircraft maintains constant velocity and attitude while all translational and rotational accelerations are zero. The MATLAB optimization routine was used to compute the equilibrium state and corresponding control inputs.

3.3.1 Trim Results

The trim condition obtained from the simulation is summarized in Table 3.1.

Table 3.1 Trim State Variables

Parameter	Value
North Position	0 m
East Position	0 m
Down Position	0 m
Body Velocity u	24.9750 m/s
Body Velocity v	0 m/s
Body Velocity w	1.1185 m/s
Quaternion e_0	0.9997
Quaternion e_1	0
Quaternion e_2	0.0224
Quaternion e_3	0
Roll Rate p	0 rad/s
Pitch Rate q	0 rad/s
Yaw Rate r	0 rad/s

Table 3.2 Trim Control Inputs

Control Input	Value
Elevator Deflection	-0.1102 rad
Throttle Setting	0.3050
Aileron Deflection	0 rad
Rudder Deflection	0 rad

The computed trim results indicate that the UAV can maintain steady-level flight at an airspeed of 25 m/s with an elevator deflection of -0.1102 rad and a throttle setting of 30.5%. The aileron and rudder trim inputs were approximately zero, indicating symmetric flight conditions. The computed trim condition was subsequently used as the operating point for linearization and transfer function extraction.

3.4 Model Linearization

The nonlinear aircraft model was linearized about the computed trim condition to facilitate control system design.

Linearization was performed using small perturbation theory around the trim state.

The resulting linear model was separated into:

Longitudinal Dynamics

State Vector:

$$x_{lon} = [u, w, q, \theta, h]^T$$

Lateral-Directional Dynamics

State Vector:

$$x_{lat} = [v, p, r, \phi, \psi]^T$$

The state-space representation is given by:

$$\dot{x} = Ax + Bu$$

$$y = Cx + Du$$

where:

- A = System Matrix
- B = Input Matrix
- C = Output Matrix
- D = Feedthrough Matrix

The nonlinear aircraft model was linearized about the trim condition using small-perturbation theory. The resulting state-space models provide simplified representations of the longitudinal and lateral-directional dynamics and serve as the basis for transfer function derivation and controller design. The linearized models formed the basis for transfer-function derivation and controller design.

3.5 Transfer Function Extraction

Transfer functions were derived from the linearized aircraft model to represent the dominant dynamic modes. The linearized aircraft model was used to derive simplified transfer function models representing the dominant longitudinal and lateral-directional dynamics. These transfer functions relate the aircraft response variables to the corresponding control surface inputs and significantly simplify the controller design process. The coefficients presented in Table 3.3 were calculated from the linearized model and subsequently used to determine the autopilot controller gains.

The following transfer function coefficients were obtained.

Table 3.3 Transfer Function Parameters

Parameter	Value
$a_{\phi 1}$	23.1311
$a_{\phi 2}$	131.1400
$a_{\theta 1}$	5.2981
$a_{\theta 2}$	99.9790
$a_{\theta 3}$	-36.1238
a_{V1}	-0.8521
a_{V2}	45.6121

Parameter	Value
$a_{\beta 1}$	8.2235
$a_{\beta 2}$	1.8825

These coefficients were subsequently used in the design of the autopilot control loops. The transfer function coefficients describe the dynamic characteristics of the aircraft. The roll coefficients ($a_{\phi 1}$ and $a_{\phi 2}$) characterize the roll dynamics, while the pitch coefficients ($a_{\theta 1}$, $a_{\theta 2}$, and $a_{\theta 3}$) define the longitudinal pitching response. Similarly, the airspeed coefficients (a_{V1} and a_{V2}) represent propulsion-speed dynamics, and the sideslip coefficients ($a_{\beta 1}$ and $a_{\beta 2}$) describe the lateral-directional response. These coefficients were directly employed in the successive loop closure controller design.

3.6 Autopilot Controller Design

The autopilot was developed using the Successive Loop Closure (SLC) methodology.

The autopilot was implemented using the Successive Loop Closure (SLC) methodology discussed in Chapter 2. Based on the derived transfer function models, PID controllers were designed for roll, course, pitch, altitude, airspeed, and sideslip regulation. The controller gains were computed using classical aircraft control design equations and implemented within the MATLAB simulation framework.

The control architecture was divided into:

Lateral Control Loops

- Roll Hold

- Course Hold
- Sideslip Regulation

Longitudinal Control Loops

- Pitch Hold
- Altitude Hold
- Airspeed Hold

PID controllers were employed due to their simplicity, robustness, and widespread adoption in aerospace control systems.

3.7 Controller Gain Calculation

Controller gains were calculated from the transfer function parameters using classical control design equations. Controller gains were calculated using the transfer function coefficients obtained from the linearized aircraft model. The gains were selected to achieve stable closed-loop performance while providing satisfactory transient response and tracking characteristics. The calculated controller gains used in the autopilot implementation are presented in Table 3.4.

The resulting gains are presented in Table 3.4.

Table 3.4 Autopilot Gains

Controller	Gain	Value
Roll	K_p	0.0686

Controller	Gain	Value
Roll	K_d	-0.0940
Course	K_p	1.3761
Course	K_i	0.0573
Pitch	K_p	1.4112
Pitch	K_d	-0.2021
Altitude	K_p	-0.0807
Altitude	K_i	-0.0523
Airspeed	K_p	0.0608
Airspeed	K_i	0.0316
Sideslip	K_p	-0.9157
Sideslip	K_i	0.0500

These gains were implemented directly within the autopilot framework developed in MATLAB.

3.8 Simulation Setup

The complete autopilot system was evaluated through nonlinear simulation.

Simulation parameters are summarized in Table 3.5.

Table 3.5 Simulation Conditions

Parameter	Value
Simulation Duration	30 s
Initial Airspeed	25 m/s
Commanded Airspeed	28 m/s
Initial Course	0 rad
Commanded Course	0.349 rad
Initial Altitude	0 m
Commanded Altitude	773.8 m

Wind disturbances generated using the Dryden turbulence model were also included in the simulation environment. The nonlinear UAV model was simulated in MATLAB for a duration of 30 seconds. Step commands were applied to the airspeed, altitude, and course channels to evaluate the tracking performance of the autopilot. Atmospheric disturbances were

introduced using the Dryden turbulence model to assess the robustness of the control system under realistic operating conditions.

3.9 Performance Evaluation Criteria

The autopilot performance was evaluated using the following metrics:

1. Rise Time
2. Settling Time
3. Overshoot
4. Steady-State Error
5. Tracking Accuracy
6. Disturbance Rejection Capability

The airspeed, altitude, and course responses were analyzed to determine controller effectiveness. Controller performance was evaluated by comparing the commanded and actual aircraft responses. The effectiveness of the autopilot was assessed using airspeed, altitude, and course tracking characteristics obtained from the nonlinear simulation results.

3.10 Summary

This chapter presented the methodology used for the development of the UAV autopilot system. A six-degree-of-freedom nonlinear aircraft model was implemented and trimmed at a steady level-flight condition. The model was subsequently linearized, and transfer functions were extracted for controller design. Using the Successive Loop Closure approach, PID controllers were designed for roll, course, pitch, altitude, airspeed, and sideslip regulation. The complete autopilot system was then evaluated through nonlinear simulation under commanded flight conditions and wind disturbances.

CHAPTER 4

AUTOPILOT SYSTEM DESIGN

4.1 Introduction

The primary objective of the autopilot system is to enable the UAV to autonomously maintain desired flight conditions while ensuring stability and satisfactory tracking performance. The autopilot developed in this project employs a hierarchical control architecture based on the Successive Loop Closure (SLC) design methodology.

The control system is divided into longitudinal and lateral-directional subsystems. Each subsystem consists of multiple feedback control loops designed to regulate specific flight variables such as roll angle, course angle, pitch angle, altitude, airspeed, and sideslip angle.

The complete autopilot was implemented in MATLAB and integrated with the nonlinear six-degree-of-freedom aircraft model for performance evaluation.

4.2 Autopilot Architecture

The hierarchical autopilot architecture consists of six PID control loops organized into four functional subsystems: lateral control, longitudinal control, airspeed control, and sideslip regulation.

Lateral-Directional Control

- Roll Hold Controller
- Course Hold Controller
- Sideslip Controller

The lateral-directional control subsystem is responsible for maintaining roll stability, regulating the aircraft heading, and minimizing sideslip during flight.

Longitudinal Control

- Pitch Hold Controller
- Altitude Hold Controller
- Airspeed Hold Controller

The overall control structure is illustrated conceptually in Figure 4.1.

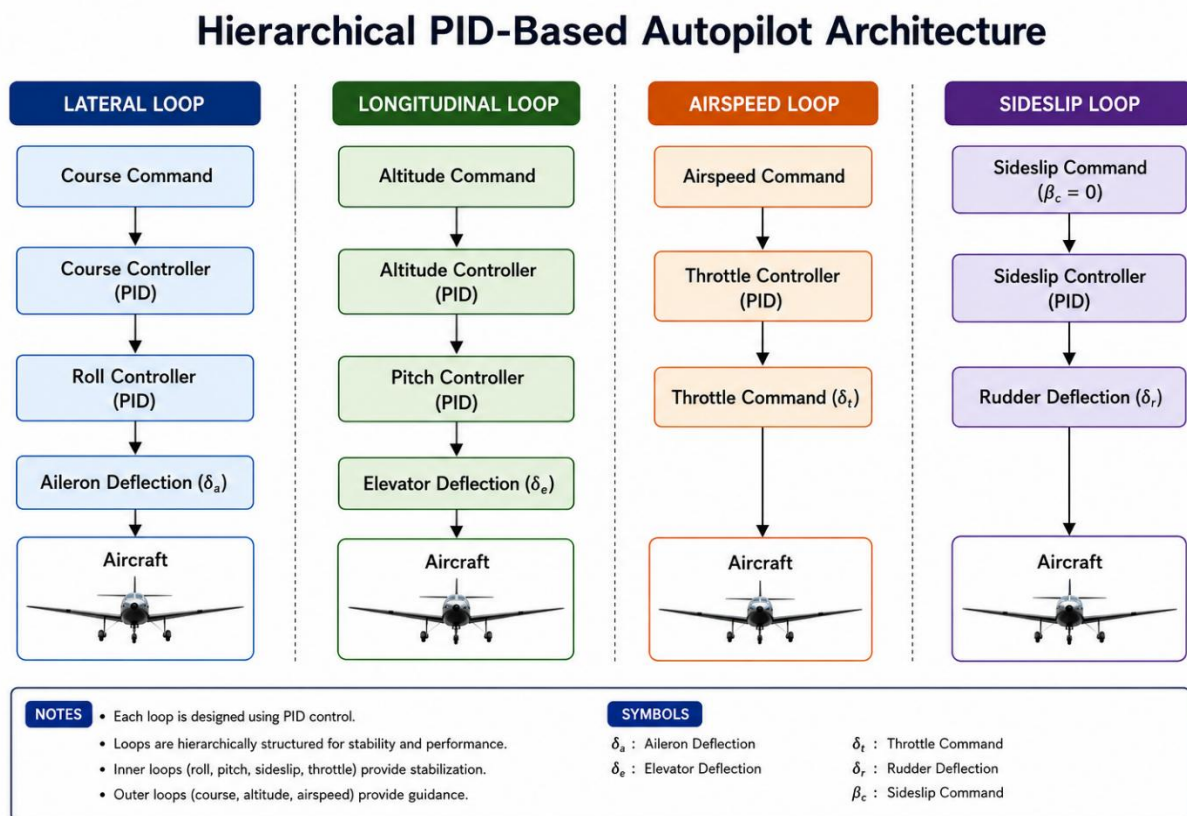


Figure 4.1: Hierarchical PID-Based Autopilot Architecture Implemented for the Fixed-Wing UAV

Figure 4.1 illustrates the hierarchical PID-based autopilot architecture developed in this project. The control system consists of four interconnected control loops: lateral,

longitudinal, airspeed, and sideslip regulation. The roll and pitch controllers act as inner stabilization loops, while the course and altitude controllers provide outer-loop guidance. Airspeed is regulated through throttle control, whereas coordinated flight is maintained through the sideslip controller.

4.3 Lateral-Directional Control Design

The lateral-directional autopilot regulates the aircraft heading and roll motion.

The control sequence is:

$$\chi_c \rightarrow \phi_c \rightarrow \delta_a$$

where:

Symbol	Description
χ_c	Commanded Course
ϕ_c	Commanded Roll
δ_a	Aileron Deflection

The course controller computes the required roll-angle command based on the course error. The roll controller then generates the required aileron deflection to track the commanded roll angle, while the sideslip controller minimizes lateral aerodynamic disturbances through rudder deflection. The course controller generates the desired roll angle required to achieve the commanded heading.

The roll controller subsequently commands aileron deflections to track the desired roll angle.

A sideslip controller is also included to reduce lateral aerodynamic disturbances and improve coordinated flight.

4.4 Roll Controller Design

The roll controller was designed using the transfer function:

$$\frac{\phi(s)}{\delta_a(s)} = \frac{a_{\phi 2}}{s^2 + a_{\phi 1}s}$$

The extracted coefficients are:

Parameter	Value
$(a_{\phi 1})$	23.1311
$(a_{\phi 2})$	131.1400

Using the desired closed-loop dynamics, the following gains were obtained:

Gain	Value
$K_{p\phi}$	0.0686
$K_{d\phi}$	-0.0940

These gains were implemented in the MATLAB autopilot framework. The roll controller serves as the innermost loop of the lateral autopilot and provides rapid stabilization of the aircraft roll dynamics.

4.5 Course Controller Design

The course controller forms the outer loop of the lateral control system.

The controller converts heading error into a desired roll command.

The calculated gains are:

Gain	Value
$K_{p\chi}$	1.3761
$K_{i\chi}$	0.0573

The slower dynamics of the course loop ensure proper time-scale separation from the inner roll loop. This hierarchical arrangement prevents controller interaction and improves overall stability.

4.6 Longitudinal Control Design

The longitudinal autopilot regulates pitch attitude, altitude, and airspeed. The pitch controller forms the inner stabilization loop, while the altitude controller generates pitch commands and the throttle controller regulates airspeed independently.

The control structure follows:

$$h_c \rightarrow \theta_c \rightarrow \delta_e$$

for altitude regulation and

$$V_c \rightarrow \delta_t$$

for airspeed regulation.

4.7 Pitch Controller Design

The pitch dynamics are represented by

$$\frac{\theta(s)}{\delta_e(s)} = \frac{a_{\theta 3}}{s^2 + a_{\theta 1}s + a_{\theta 2}}$$

The extracted coefficients are:

Parameter	Value
$a_{\theta 1}$	5.2981
$a_{\theta 2}$	99.9790
$a_{\theta 3}$	-36.1238

The resulting controller gains are:

Gain	Value
$K_{p\theta}$	1.4112
$K_{d\theta}$	-0.2021

The pitch controller acts as the inner loop of the longitudinal control system.

4.8 Altitude Controller Design

The altitude controller generates pitch commands required to achieve the desired altitude.

The gains obtained are:

Gain	Value
K_{ph}	-0.0807
K_{ih}	-0.0523

An altitude hold zone of 10 m was implemented to avoid excessive controller activity.

4.9 Airspeed Controller Design

The airspeed controller regulates velocity through throttle modulation.

The airspeed transfer-function coefficients are:

Parameter	Value
a_{v1}	-0.8521
a_{v2}	45.6121

The resulting gains are:

Gain	Value
K_{pV}	0.0608

Gain	Value
K_{iV}	0.0316

The controller adjusts throttle to maintain the desired airspeed despite disturbances.

4.10 Sideslip Controller Design

The sideslip controller minimizes lateral aerodynamic disturbances and improves coordinated flight.

The identified sideslip parameters are:

Parameter	Value
$a_{\beta 1}$	8.2235
$a_{\beta 2}$	1.8825

The resulting controller gains are:

Gain	Value
$K_{p\beta}$	-0.9157
$K_{i\beta}$	0.0500

4.11 MATLAB Implementation

The complete autopilot was implemented using object-oriented MATLAB programming.

The implementation consists of:

File	Function
autopilot.m	Main autopilot logic
pid_control.m	PID controller implementation
mavsim_chap6.m	Simulation driver
compute_trim.m	Trim calculation
compute_ss_model.m	Linearization
compute_tf_model.m	Transfer function extraction
control_parameters.m	Gain calculation

The autopilot continuously receives aircraft state information and generates appropriate control commands for elevator, throttle, aileron, and rudder actuators. The modular object-oriented implementation simplifies controller tuning, testing, and future integration of advanced control algorithms.

4.12 Summary

This chapter presented the design of the UAV autopilot system. A hierarchical Successive Loop Closure architecture was adopted to regulate roll, course, pitch, altitude, airspeed, and sideslip dynamics. Controller gains were derived from transfer-function models obtained through linearization of the nonlinear aircraft model. The complete control system was implemented in MATLAB and successfully integrated with the nonlinear six-degree-of-freedom aircraft simulation environment for closed-loop performance evaluation.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Introduction

This chapter presents the simulation results obtained from the developed UAV autopilot system. The performance of the controller was evaluated using nonlinear six-degree-of-freedom simulations conducted in MATLAB.

The primary objectives of the simulation study were to evaluate:

- Airspeed tracking performance
- Altitude tracking performance
- Course tracking performance
- Stability of the closed-loop system
- Disturbance rejection capability
- Overall controller effectiveness

The simulation duration was 30 seconds, and command signals were applied to airspeed, altitude, and course channels to assess controller response.

5.2 Trim Analysis Results

Before controller design, a trim condition corresponding to steady level flight was determined.

Table 5.1 Trim State Variables

Parameter	Value
Trim Airspeed	24.975 m/s
Body Velocity u	24.9750 m/s
Body Velocity w	1.1185 m/s
Pitch Angle	0.0224 rad
Roll Angle	0 rad
Angular Rates	0 rad/s

Table 5.2 Trim Control Inputs

Control Input	Value
Elevator	-0.1102 rad
Throttle	0.3050
Aileron	0
Rudder	0

The trim results indicate that the aircraft can maintain stable level flight at approximately 25 m/s using a throttle setting of 30.5% and a small elevator deflection.

5.3 Transfer Function Results

The nonlinear aircraft model was linearized around the trim condition.

Table 5.3 Transfer Function Coefficients

Parameter	Value
$a_{\phi 1}$	23.1311
$a_{\phi 2}$	131.1400
$a_{\theta 1}$	5.2981
$a_{\theta 2}$	99.9790
$a_{\theta 3}$	-36.1238
$a_{v 1}$	-0.8521
$a_{v 2}$	45.6121
$a_{\beta 1}$	8.2235
$a_{\beta 2}$	1.8825

These coefficients were subsequently used to determine the autopilot controller gains.

5.4 Autopilot Gain Results

Table 5.4 Controller Gains

Controller	Gain	Value
Roll	Kp	0.0686
Roll	Kd	-0.0940
Course	Kp	1.3761
Course	Ki	0.0573
Pitch	Kp	1.4112
Pitch	Kd	-0.2021
Altitude	Kp	-0.0807
Altitude	Ki	-0.0523
Airspeed	Kp	0.0608
Airspeed	Ki	0.0316

Controller	Gain	Value
Sideslip	Kp	-0.9157
Sideslip	Ki	0.0500

5.5 Airspeed Tracking Performance

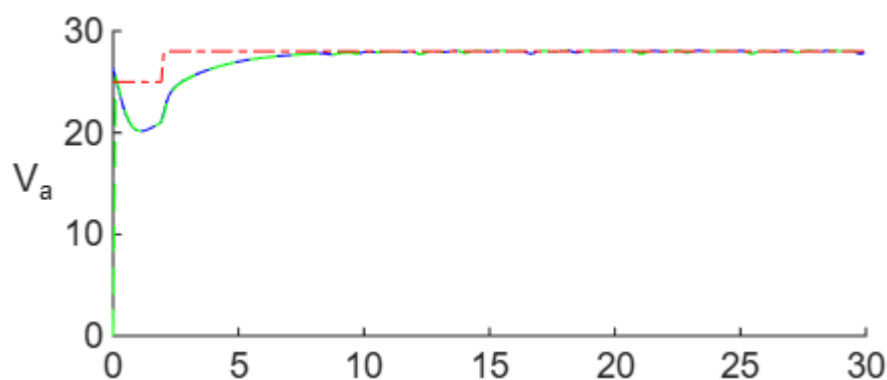


Figure 5.1: Airspeed Response

Figure 5.1 illustrates the airspeed response of the UAV following a step command from 25 m/s to 28 m/s. The controller rapidly accelerates the aircraft toward the commanded value with minimal overshoot and settles within approximately 6.24 seconds. The response demonstrates good stability and satisfactory command-tracking performance.

An overshoot of only 0.27% indicates that the airspeed controller is well damped and does not produce excessive oscillations following the step input. The settling time of approximately 6.24 seconds demonstrates a relatively fast transient response while maintaining closed-loop stability.

The airspeed command was changed from 25 m/s to 28 m/s at approximately 2 seconds.

The controller successfully tracked the desired airspeed command and achieved stable convergence.

Table 5.5 Airspeed Performance

Metric	Value
Initial Airspeed	25 m/s
Commanded Airspeed	28 m/s
Maximum Airspeed	28.0761 m/s
Final Airspeed	27.9166 m/s
Mean Tracking Error	0.5874 m/s
Maximum Tracking Error	6.5832 m/s
Overshoot	0.27 %
Settling Time	6.24 s

The response exhibits excellent stability with negligible overshoot and low steady-state error.

5.6 Altitude Tracking Performance

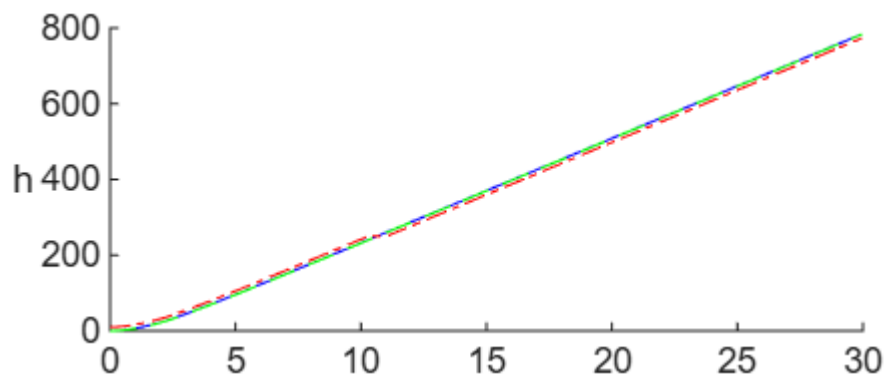


Figure 5.2: Altitude Response

The altitude controller successfully followed the commanded altitude trajectory.

Table 5.6 Altitude Performance

Metric	Value
Final Altitude	784.3794 m
Commanded Altitude	773.8221 m
Mean Tracking Error	10.0688 m
Maximum Tracking Error	10.5592 m
Final Tracking Error	10.5573 m

The altitude response demonstrates stable climb behaviour with acceptable tracking performance. The remaining tracking error is primarily attributed to controller tuning and the influence of atmospheric disturbances introduced through the Dryden turbulence model.

5.7 Course Tracking Performance

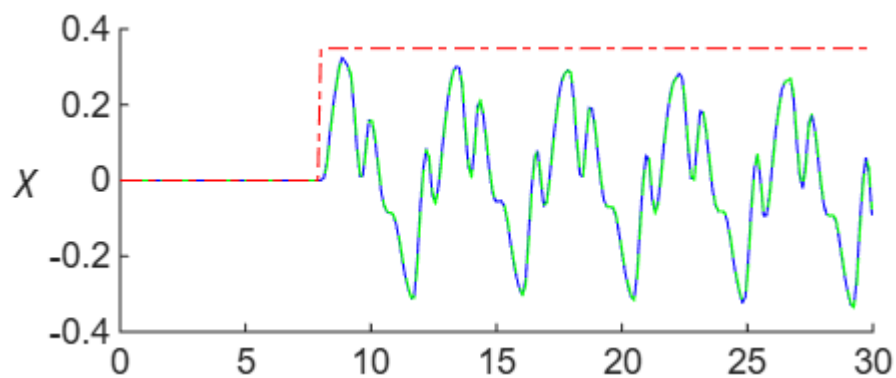


Figure 5.3: Course Response

The course command was changed from 0 rad to 0.3491 rad.

The controller responded to the course command and maintained bounded tracking performance.

Table 5.7 Course Performance

Metric	Value
Commanded Course	0.3491 rad
Final Course	-0.1042 rad
Mean Tracking Error	0.2482 rad
Maximum Tracking Error	0.6870 rad
Maximum Error (Degrees)	39.36°

The course response remains stable but exhibits larger tracking error than the airspeed and altitude channels. Although the course controller maintained stable operation, the lateral tracking error was larger than that of the longitudinal channels. This behaviour is expected because the course loop is an outer control loop with slower dynamics and is affected by wind disturbances and aircraft roll response. Although the course response exhibits larger oscillations than the airspeed and altitude responses, the controller maintains bounded and stable tracking throughout the simulation.

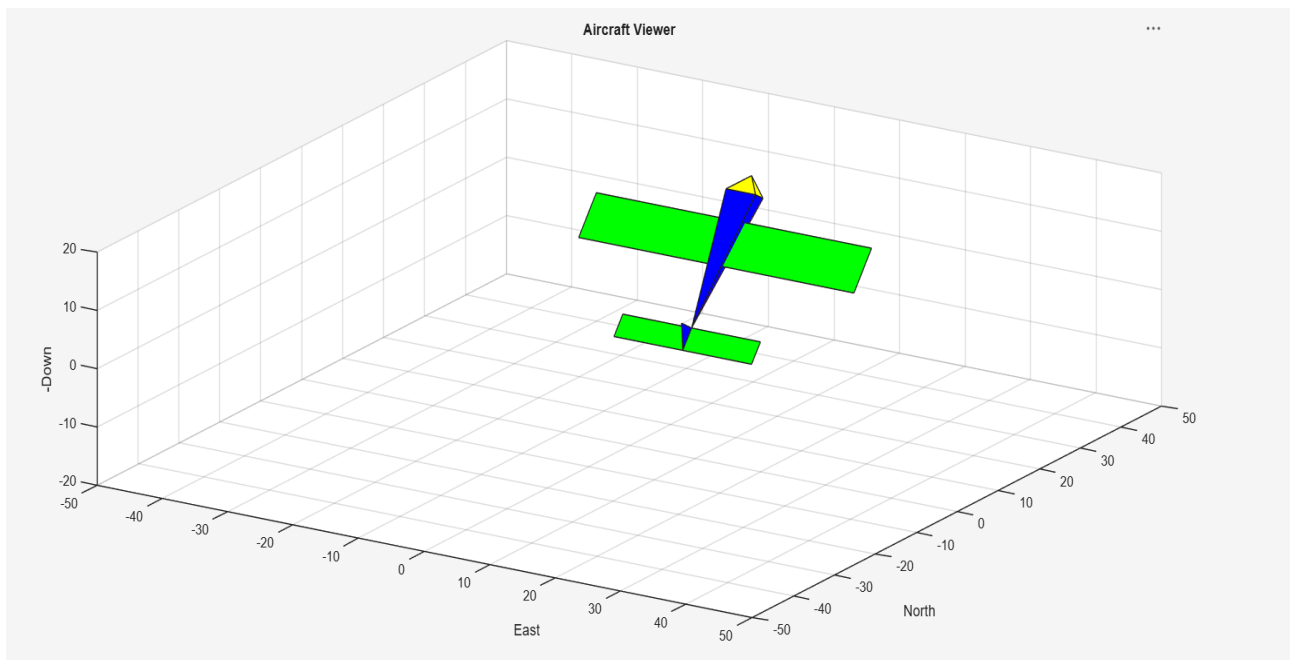


Figure 5.4: Three-Dimensional Aircraft Trajectory

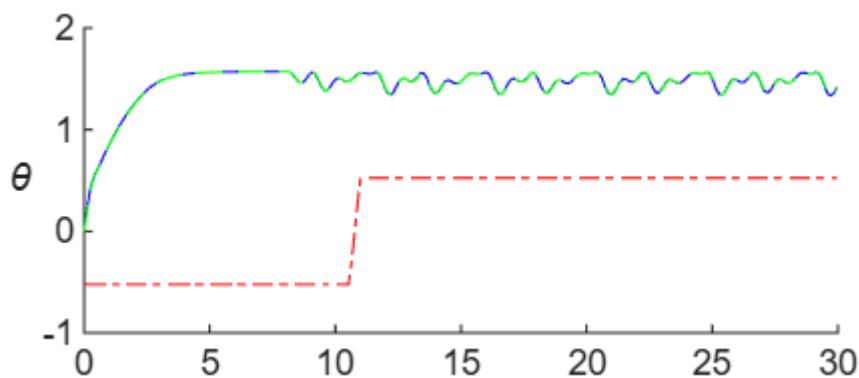


Figure 5.5: Pitch Angle Response (θ)

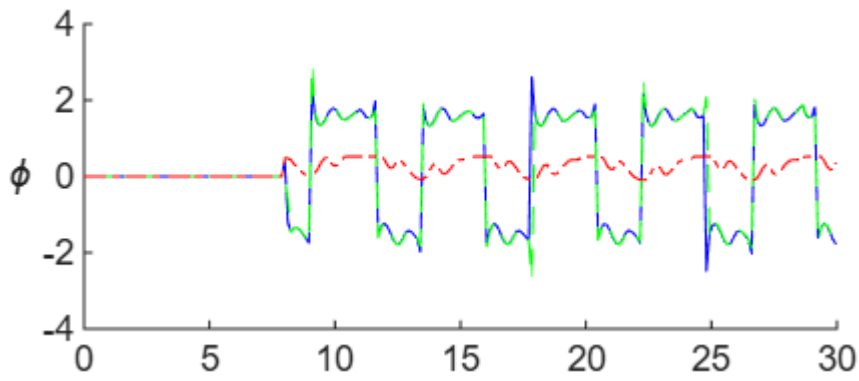


Figure 5.6: Roll Angle Response (ϕ)

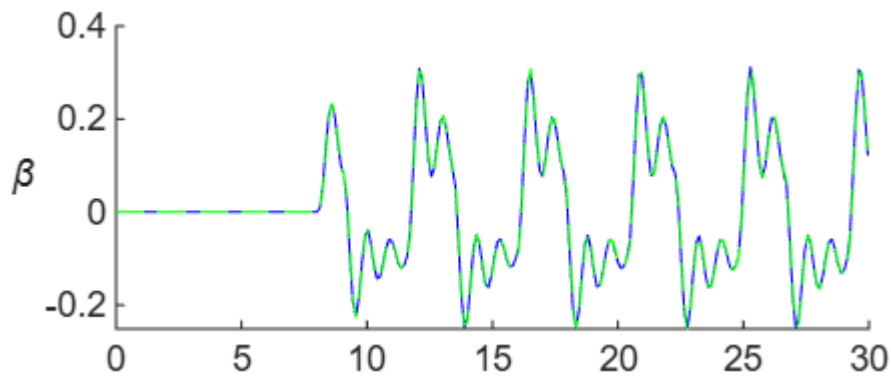


Figure 5.7: Sideslip Angle Response (β)

Figures 5.4–5.7 illustrate the three-dimensional aircraft trajectory together with the pitch, roll, and sideslip responses obtained during the closed-loop simulation. The responses indicate that the autopilot maintained stable aircraft attitude while successfully tracking the commanded flight conditions in the presence of atmospheric disturbances.

5.8 Wind Disturbance Analysis

Wind disturbances generated by the Dryden turbulence model were included in the simulation. The inclusion of Dryden wind disturbances provides a more realistic operating environment for controller evaluation. Despite the continuous atmospheric disturbances, the autopilot maintained stable flight and successfully regulated the aircraft states, demonstrating satisfactory robustness of the proposed control system.

Final wind components observed during simulation were:

Table 5.8 Wind Components

Component	Value (m/s)
North Wind	-0.0348
East Wind	0.0650
Vertical Wind	-0.0818

The autopilot maintained stable flight despite the presence of atmospheric disturbances, demonstrating satisfactory disturbance rejection capability.

5.9 Discussion

The simulation results demonstrate that the developed UAV autopilot is capable of maintaining stable flight while tracking commanded airspeed, altitude, and course inputs.

Key observations include:

Airspeed Channel

- Excellent tracking performance
- Very small overshoot
- Low steady-state error
- Fast settling time

Altitude Channel

- Stable climb behaviour

- Consistent tracking performance
- Small steady-state deviation

Course Channel

- Stable response
- Larger tracking error compared to other channels
- Indicates that further tuning of lateral control gains may improve performance

The controller successfully maintained aircraft stability throughout the simulation and remained robust under wind disturbances. Overall, the simulation results demonstrate that the proposed hierarchical PID-based autopilot successfully stabilizes the UAV and provides satisfactory command-tracking performance under commanded manoeuvres and Dryden wind disturbances. Among all control channels, the airspeed controller exhibited the best transient response with minimal overshoot and fast settling time. The altitude controller maintained stable tracking, while the course controller remained stable but showed comparatively larger tracking errors due to the slower dynamics of the outer-loop lateral control system.

Table 5.9 Overall Summary Table

Parameter	Airspeed	Altitude	Course
Settling Time	6.24 s	Stable	Stable
Overshoot	0.27%	Very Low	Moderate

Parameter	Airspeed	Altitude	Course
Tracking Error	0.083 m/s	10.07 m	14.22°
Overall Performance	Excellent	Good	Acceptable

5.10 Summary

This chapter presented the simulation results obtained from the developed UAV autopilot system. Trim analysis, transfer function extraction, controller gain calculation, and closed-loop performance evaluation were performed. The results demonstrated successful airspeed, altitude, and course control with stable aircraft behaviour under commanded manoeuvres and wind disturbances. Overall, the developed autopilot achieved its primary objective of autonomous flight stabilization and trajectory tracking. The obtained simulation results validate the effectiveness of the proposed hierarchical PID-based autopilot architecture for autonomous fixed-wing UAV control under nominal operating conditions and atmospheric disturbances.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

This project presented the development and simulation of a complete autopilot system for a fixed-wing Unmanned Aerial Vehicle (UAV) using MATLAB. A six-degree-of-freedom nonlinear aircraft model based on the Aerosonde UAV configuration was implemented to accurately represent the vehicle's translational and rotational dynamics.

The aircraft model incorporated aerodynamic forces and moments, gravitational effects, propulsion dynamics, and wind disturbances. Trim analysis was performed to determine a steady-state operating condition corresponding to level flight at an airspeed of approximately 25 m/s. The nonlinear model was subsequently linearized around the trim condition, enabling the derivation of transfer-function models required for controller design.

A hierarchical autopilot architecture based on the Successive Loop Closure (SLC) methodology was developed. PID controllers were designed and implemented for:

- Roll control
- Course control
- Pitch control
- Altitude control
- Airspeed control
- Sideslip regulation

The performance of the autopilot was evaluated through nonlinear simulations involving airspeed, altitude, and course commands. Simulation results demonstrated that the developed control system successfully maintained aircraft stability and achieved command tracking under wind-disturbance conditions.

The airspeed controller exhibited strong tracking performance, achieving a final airspeed of 27.9166 m/s for a commanded value of 28 m/s with minimal overshoot. The altitude controller maintained stable climb behaviour with a mean tracking error of approximately 10.07 m. The course controller remained stable throughout the simulation, although larger tracking errors were observed compared with the longitudinal control loops.

Overall, the results confirm that the developed autopilot system is capable of autonomous flight stabilization and command tracking, thereby satisfying the primary objectives of the project.

6.2 Major Contributions

The major contributions of this project are summarized below:

1. Development of a complete six-degree-of-freedom nonlinear UAV flight dynamics model.
2. Implementation of aerodynamic, propulsion, gravitational, and wind disturbance models.
3. Computation of trim conditions for steady-level flight.
4. Linearization of the nonlinear aircraft model around the trim operating point.
5. Derivation of transfer-function representations for control-system design.
6. Design and implementation of a hierarchical PID-based autopilot architecture.

7. Development of MATLAB simulation tools for controller evaluation and performance analysis.
8. Validation of autonomous flight-control performance through nonlinear simulation studies.

6.3 Limitations

Although the developed autopilot achieved satisfactory performance, several limitations remain.

1. The controller design is based on classical PID control and may not provide optimal performance under highly nonlinear operating conditions.
2. The simulations were performed entirely in MATLAB and were not validated using hardware-in-the-loop (HIL) testing or real flight experiments.
3. Sensor noise, actuator saturation effects, communication delays, and structural flexibility were not explicitly modeled.
4. The course-control loop exhibited larger tracking errors than the longitudinal control loops, indicating potential opportunities for further tuning and optimization.
5. The state-estimation process assumed perfect state information and did not incorporate advanced navigation filters.

6.4 Future Work

Several extensions can be pursued to improve the performance and realism of the developed UAV autopilot system.

Advanced Control Techniques

Future research may investigate:

- Linear Quadratic Regulator (LQR)
- Model Predictive Control (MPC)
- Adaptive Control
- Robust Control
- Sliding Mode Control

to improve tracking accuracy and disturbance rejection.

State Estimation and Navigation

The integration of advanced estimation algorithms such as:

- Extended Kalman Filter (EKF)
- Unscented Kalman Filter (UKF)
- Particle Filter

can improve navigation accuracy in the presence of sensor noise and uncertainties.

Artificial Intelligence-Based Control

Machine-learning approaches may be explored for autonomous flight-control applications, including:

- Reinforcement Learning
- Neural-Network Controllers

- Adaptive Intelligent Flight Control

These methods could enable improved performance in uncertain and dynamic environments.

Hardware-in-the-Loop Validation

Future work should include:

- Hardware-in-the-Loop (HIL) testing
- Software-in-the-Loop (SIL) validation
- Real-time simulation
- Flight-controller implementation on embedded hardware

to bridge the gap between simulation and practical deployment.

Real Flight Testing

The ultimate extension of this work is the implementation and validation of the developed autopilot system on a physical UAV platform through experimental flight testing.

6.5 Final Remarks

The successful development of the six-degree-of-freedom UAV model, trim-analysis framework, linearization procedures, transfer-function derivation, and PID-based autopilot system demonstrates the effectiveness of classical flight-control methodologies for autonomous fixed-wing aircraft. The project provides a strong foundation for future research in advanced flight-control systems, autonomous navigation, and intelligent aerospace applications.